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What is K-Means Clustering?

K-Means is a method used to **divide data into K groups (clusters)**.

👉 "K" means number of groups you want.

📌 Example:

If $K = 3 \rightarrow$ data will be divided into 3 clusters



How K-Means Works (Step-by-Step)

Step 1:

Choose number of clusters (K)

Step 2:

Select random points as **centroids** (center of clusters)

Step 3:

Assign each data point to the **nearest centroid**

Step 4:

Update centroid (calculate new center)

Step 5:

Repeat steps 3 & 4 until no change



Simple Understanding

👉 It keeps grouping data until:

- Each group has similar points
 - Centers (centroids) become stable
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Advantages

- Easy to understand
 - Fast for large data
 - Works well for simple clusters
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Disadvantages

- Need to choose K manually
 - Not good for irregular shapes
 - Sensitive to noise
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Real-Life Example

👉 Customer Segmentation

- Group customers based on spending habits
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Conclusion (Write in exam)

K-Means is a simple and efficient clustering algorithm that divides data into K groups by assigning data points to the nearest centroid and updating cluster centers until stability is achieved.



What is Hierarchical Clustering?

Hierarchical Clustering is a method of grouping data that **creates a tree-like structure (called dendrogram)**.



It shows how data points are grouped step by step.



Simple idea:

“Keep joining or dividing groups until clusters are formed.”



Types of Hierarchical Clustering

1. Agglomerative (Bottom → Up)

- Start with each data point as a **separate cluster**
- Merge closest clusters step by step
- Continue until one big cluster is formed



Most commonly used

2. Divisive (Top → Down)

- Start with all data in **one cluster**
- Split into smaller clusters
- Continue dividing

Here's a **very simple real-life example of Hierarchical Clustering** you can write in exam 🙋

Real-Life Example: Student Grouping

Situation:

A teacher wants to group students based on their marks.

Marks:

- A → 90
 - B → 88
 - C → 85
 - D → 40
 - E → 42
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How Hierarchical Clustering Works:

👉 Step 1:

Each student is a separate group

👉 Step 2:

Closest students are grouped:

- A, B, C → High marks group
- D, E → Low marks group

👉 Step 3:

These groups can further combine into one big group

Final Groups:

- Cluster 1 → High performers (A, B, C)
 - Cluster 2 → Low performers (D, E)
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Simple Understanding

👉 Students with similar marks are grouped together automatically

Write in Exam (Short)

Hierarchical clustering can be used to group students based on marks. Initially, each student is treated as a separate cluster, and similar students are merged

step by step. Finally, groups like high performers and low performers are formed.

If you want, I can give you **more real-life examples** (business, medical, shopping)



Situation:

A shopping mall wants to **group customers** based on:

- Income
 - Spending habits
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How K-Means Works Here

 Step 1:

Choose $K = 3$ (3 groups)

 Step 2:

System creates 3 clusters like:

 **Cluster 1:**

- High income, high spending
-  Premium customers

 **Cluster 2:**

- Medium income, medium spending

👉 Regular customers

👤👤👤 Cluster 3:

- Low income, low spending

👉 Budget customers

Result

Customers are divided into **3 clear groups**

👉 Company can:

- Give offers to each group
- Improve marketing strategy

What is DBSCAN?

DBSCAN is a clustering method that **groups data based on dense areas (crowded points)**.

👉 Full form: **Density-Based Spatial Clustering of Applications with Noise**

🔴 Simple idea:

“Points that are close and crowded form a cluster, and far points are noise.”

How DBSCAN Works (Easy)

DBSCAN uses 2 important things:

1. Epsilon (ϵ)

👉 Distance range (how close points should be)

2. MinPts

👉 Minimum number of points to form a cluster



Types of Points

- **Core Point** → Many neighbors (dense area)
 - **Border Point** → Few neighbors (edge of cluster)
 - **Noise Point** → No neighbors (outlier)
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Real-Life Example: Fraud Detection

👉 In banking:

- Normal transactions → form clusters
 - Unusual transactions → marked as **noise (fraud)**
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Why DBSCAN is Useful

- Finds clusters of **any shape**
 - Detects **outliers (noise)**
 - No need to choose number of clusters (K)
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Advantages

- Handles noise well
- Works for irregular shapes



Disadvantages

- Choosing ϵ and MinPts is difficult
- Not good for varying density



Write in Exam (Short)

DBSCAN is a density-based clustering algorithm that groups data points based on dense regions and identifies outliers as noise. It uses parameters like epsilon and minimum points to form clusters and is useful for detecting irregular shapes and anomalies.